Problem 6.77: a step-by-step attack plan

Integral means of univalent polynomials

The problem

Let $\mathcal{P}_n$ be the class of polynomials $p(z) = z + a_2 z^2 + \cdots + a_n z^n$ univalent in $\mathbb{D}$. Determine

$$M_n(q) = \max_{p \in \mathcal{P}_n} J_q(p), \qquad J_q(p) = \int_0^{2\pi} |p(e^{it})|^q \, dt, \qquad 0 < q < \infty.$$

Strategy in one line. Explore numerically to find the extremal polynomial and test the natural conjecture (Suffridge polynomials), then prove the first nontrivial case ($n = 3$) rigorously, starting with $q = 2$ where the objective is quadratic, then all q.

Division of labour. The MacBook (M4, 24 GB) does all development, small-n exploration, exact/symbolic work and Lean. RunPod is for embarrassingly parallel jobs: large multi-start sweeps ($n \geq 5$, many q) and branch-and-bound certificate runs. Rigorous verification is CPU work; GPUs are optional.

Status of this document. Everything labelled *Lemma* below is proved here (short proofs). Everything labelled *numerical* is floating-point evidence, not proof. Literature items marked "check" must be confirmed before you rely on them.

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1. Mathematical groundwork (proved facts you can use immediately)

Lemma 1.1: The maximum exists

$\mathcal{P}_n$ is a compact subset of $\mathbb{C}^{n-1}$ (coefficients $a_2, \dots, a_n$), so $M_n(q)$ is attained.

Proof. Bounded: every $p \in \mathcal{P}_n$ is univalent with $p(0) = 0$, $p'(0) = 1$, so $|a_k| \le k$ (de Branges). Closed: if $p_j \to p$ coefficientwise then $p_j \to p$ locally uniformly on $\mathbb{D}$; by Hurwitz's theorem a locally uniform limit of univalent functions is univalent or constant, and $p'(0) = 1$ excludes constant. J_q is continuous in the coefficients. □

Lemma 1.2: The maximizer lies on the boundary of $\mathcal{P}_n$

J_q is plurisubharmonic in $(a_2, \dots, a_n)$. Hence its maximum over the compact set $\mathcal{P}_n$ is attained on $\partial\mathcal{P}_n$, and through a maximizer there is no nonconstant analytic disc in $\mathcal{P}_n$ along which J_q is nonconstant.

Proof. For fixed t, $(a_2, \dots, a_n) \mapsto p(e^{it})$ is affine, so $\log|p(e^{it})|$ is plurisubharmonic and so is $|p(e^{it})|^q = e^{q \log|p(e^{it})|}$. Integrating in t preserves plurisubharmonicity. Restricted to any complex line, J_q is subharmonic, so it obeys the maximum principle. □

Lemma 1.3: The integrand is smooth

If $p \in \mathcal{P}_n$ then $p(e^{it}) \neq 0$ for all t. Hence $t \mapsto |p(e^{it})|^q$ is real-analytic and the trapezoid rule converges exponentially fast.

Proof. First, $p(\partial\mathbb{D}) \cap p(\mathbb{D}) = \emptyset$. Suppose $p(e^{it}) = p(w)$ with $w \in \mathbb{D}$, and take a small disc $U \ni w$ whose closure misses the ray $\{re^{it}\}$ near $r = 1$. Since $p(U)$ is an open set containing $p(e^{it})$, continuity gives $p(re^{it}) \in p(U)$ for r close to 1. So $p(re^{it}) = p(u)$ with $u \in U$, $u \neq re^{it}$, contradicting injectivity on $\mathbb{D}$. Since $0 = p(0) \in p(\mathbb{D})$, p has no zeros on $\partial\mathbb{D}$. Then $|p|^q = \exp(q \log|p|)$ is real-analytic in t. □

Lemma 1.4: Last coefficient

For $p \in \mathcal{P}_n$: $|a_n| \le 1/n$.

Proof. $p'(z) = na_n \prod_{j=1}^{n-1}(z - \zeta_j)$ and $p' \neq 0$ in $\mathbb{D}$, so $|\zeta_j| \ge 1$. Then $1 = |p'(0)| = n|a_n| \prod |\zeta_j| \ge n|a_n|$. □

Dieudonné's criterion (classical; check the statement in Duren, *Univalent Functions*, or Sheil-Small, *Complex Polynomials*). $p \in \mathcal{P}_n$ if and only if for every $0 < \theta < \pi/2$

$$D_\theta(z) = \sum_{k=1}^{n} a_k \frac{\sin k\theta}{\sin\theta} z^{k-1} \neq 0 \qquad (|z| < 1).$$

Lemma 1.5: Dilation and an explicit boundary map

For $p \in \mathbb{C}[z]$ with $a_1 = 1$ put $p_r(z) = p(rz)/r$ (coefficients $a_k r^{k-1}$) and

$$R(p) = \inf_{0<\theta<\pi/2} \min\{|z| : D_\theta(z) = 0\}.$$

Then $p_r \in \mathcal{P}_n \iff r \le R(p)$. In particular $p_{R(p)} \in \partial\mathcal{P}_n$, and $\partial\mathcal{P}_n = \{p : R(p) = 1\}$.

Proof sketch. The Dieudonné polynomial of p_r is $D_\theta(rz)$, which is nonzero on $|z| < 1$ iff D_θ is nonzero on $|w| < r$. Taking all θ gives the equivalence. Continuity of roots in the coefficients suggests that $R > 1$ is an open condition (interior) and $R = 1$ is the boundary. Making this rigorous needs care at $\theta \to 0$ and $\theta \to \pi/2$, where the degree of D_θ can drop; write this out before relying on it. □

Consequence for computation. Maximizing over $\mathcal{P}_n$ becomes an *unconstrained* problem: optimize $c \mapsto J_q\big(p^{(c)}_{R(p^{(c)})}\big)$ over arbitrary coefficient vectors c. This is what the starter code does.

Two reformulations worth knowing.

- $q = 2$ *is quadratic:* by Parseval, $J_2(p) = 2\pi\big(1 + |a_2|^2 + \cdots + |a_n|^2\big)$.
- *All q at once (majorization).* If $\log|p(e^{it})|$ is majorized by $\log|p^\star(e^{it})|$ in the Hardy–Littlewood–Pólya sense (equivalently, Baernstein's star function of p is dominated by that of $p^\star$), then $\int \Phi(\log|p|) \le \int \Phi(\log|p^\star|)$ for every convex increasing Φ, in particular $\Phi(x) = e^{qx}$ for every $q > 0$. This is the mechanism behind Baernstein's theorem for the Koebe function and the natural route to an all-q result.

The candidate. The Suffridge polynomial

$$S_n(z) = \sum_{k=1}^{n} \frac{n-k+1}{n} \cdot \frac{\sin\frac{k\pi}{n+1}}{\sin\frac{\pi}{n+1}}\, z^k$$

is univalent and lies on $\partial\mathcal{P}_n$ (numerically $R(S_n) = 1$ for $n \le 5$). For $n = 2$ it is $z + \frac{1}{2}z^2$, the proven maximizer for every q (subharmonicity argument). Caution: Suffridge polynomials are extremal for some problems over $\mathcal{P}_n$ but not for others (check the Dmitrishin–Smorodin–Stokolos papers on the Koebe radius of univalent polynomials), so do not assume they win here. Step 3 tests exactly this.

2. Step 0: literature and statement check (2–4 days, MacBook)

Goal

Make sure the problem is still open in the form you will attack, and collect every known tool.

1. Read the *Update* paragraph under 6.77 in Hayman–Lingham (2019 edition). Note who posed it and any partial results.
2. Search (MathSciNet, Google Scholar, an AI deep-research tool) for papers citing:

- Suffridge, "On univalent polynomials" (late 1960s) — origin of S_n;
- Brannan (1967) and Cowling–Royster (late 1960s) on coefficient regions of univalent polynomials of small degree;
- Dimitrov and coauthors; Dmitrishin–Smorodin–Stokolos (extremal univalent polynomials);
- Baernstein (1974), integral means and circular symmetrization.

Check each citation yourself; record exact bibliographic data.

3. Search specifically for "integral means", "Hardy norm", "H^p", "area" of univalent polynomials. If the case $n = 3$ or $q = 2$ is done, adjust the target.

3. Step 1: environment (1 day)

MacBook (Apple silicon).

- Python 3.12 with `numpy scipy matplotlib mpmath python-flint numba` (`python-flint` gives Arb ball arithmetic for rigorous enclosures).
- Optional: SageMath (via conda-forge) for exact/semialgebraic computation; Lean 4 + Mathlib via `elan` for formal checks of small lemmas.
- A git repository holding code, a `results/` folder (JSON with seeds, parameters, timestamps) and a lab notebook.

RunPod.

- Default: a *CPU* pod with many vCPUs; run sweeps with Python `multiprocessing`, one process per start or per box.
- Optional: a GPU pod only for very large sweeps, using batched root-finding (eigenvalues of stacked companion matrices with PyTorch/JAX).
- Workflow: clone the repo, run inside `tmux`, write results as JSON, `rsync` back, shut the pod down. Check current pricing; the whole project should cost tens to low hundreds of dollars.

4. Step 2: numerical toolkit (2–3 days, MacBook)

The starter file `p677.py` (delivered with this plan) already contains:

- `suffridge(n)`: coefficients of S_n;
- `dieudonne_radius(a)`: $R(p)$ from Lemma 1.5;
- `project_to_boundary(a)`: the boundary point $p_{R(p)}$;
- `J(a,q)`: trapezoid rule (spectrally accurate by Lemma 1.3);
- `maximize(n,q)`: multi-start Nelder–Mead over boundary polynomials.

Unit tests that pass: $z + az^2$ is univalent iff $|a| \le \frac{1}{2}$; $R(S_n) = 1$ for $n = 2, \dots, 5$; $z + 2z^2 + 3z^3$ has $R = \frac{1}{3}$.

Improvements to make first.

1. Vectorize $R(p)$: build all θ companion matrices at once and call `np.linalg.eigvals`; refine the minimizing θ by golden-section search (the grid minimum is only approximate).

2. Add a gradient-based optimizer (BFGS on the boundary parametrization) for polishing.
3. Add the $q = \infty$ functional $\max_t |p(e^{it})|$ (by rotation, the maximum of $|p(1)|$).

5. Step 3: exploration — is Suffridge the answer? (1–2 weeks)

Goal

For $n = 3, \ldots, 8$ and $q \in \{0.25, 0.5, 1, 2, 3, 4, 8\}$ plus $q = \infty$: find $M_n(q)$ numerically and the maximizer.

Preliminary data (already computed, numerical, 6 random starts).

n	q	best found J_q	$J_q(S_3)$
3	2	12.566371	12.566371 $(= 4\pi)$
3	1	7.703972	7.703972

In both runs the optimizer converged to $S_3 = z + \frac{2\sqrt{2}}{3}z^2 + \frac{1}{3}z^3$ (up to rotation). Note $\sum |a_k|^2 = 2$ for S_3, a suspiciously clean value worth explaining.

Protocol.

1. MacBook: $n = 3, 4$, all q, 50 starts each.
2. RunPod CPU pod: $n = 5, \ldots, 8$, all q, 500–2000 starts each, in parallel.
3. For every maximizer record: coefficients (after rotating $a_2 > 0$), whether they are real, the location of the critical points (zeros of p'), the θ attaining $R(p) = 1$, and $J_q(S_n) - M_n(q)$.
4. Decision point:
   - If some p beats S_n by more than numerical error: you have a counterexample to the natural conjecture. Certify it (Step 6) and write it up — that is a result.
   - If S_n always wins: you have strong evidence; move to proofs.
5. Also test majorization numerically: for random $p \in \partial\mathcal{P}_n$, compare decreasing rearrangements of $\log |p(e^{it})|$ and $\log |S_n(e^{it})|$. If majorization holds in all tests, the all-q route of Section 1 is promising.

6. Step 4: structure and conjectures (ongoing; MacBook + LLM API)

Use a frontier model through its API as a *generator*: propose lemmas, write code, search literature. Treat every output as unverified.

- Candidate lemmas to test: the maximizer has real coefficients after rotation; all critical points of the maximizer lie on $|z| = 1$ (true for S_n); the maximizer is the same for all q; $p \mapsto J_q(p)$ restricted to $\partial\mathcal{P}_n$ has a unique critical point modulo rotation.
- Ask why $\sum |a_k|^2 = 2$ for S_3; compute $\sum |a_k|^2$ for S_n in closed form and compare with the maximum found numerically.
- Look for a polynomial analogue of circular symmetrization; understand why Baernstein's proof does not directly apply (symmetrization does not preserve the polynomial class).

7. Step 5: rigorous proof of $n = 3$, $q = 2$ (2–4 weeks)

Goal

Prove $\max_{p\in\mathcal{P}_3}(1 + |a_2|^2 + |a_3|^2) = 2$, attained exactly by rotations of S_3 (if Step 3 confirms this).

Explicit description of $\mathcal{P}_3$. With $x = \cos\theta \in (0, 1)$, $\frac{\sin 2\theta}{\sin\theta} = 2x$ and $\frac{\sin 3\theta}{\sin\theta} = 4x^2 - 1$, so

$$D_\theta(z) = 1 + b(x)z + c(x)z^2, \qquad b(x) = 2a_2x, \quad c(x) = a_3(4x^2 - 1).$$

D_θ has no zeros in $|z| < 1$ iff the reversed quadratic $z^2 + bz + c$ has both zeros in the closed disc. The Schur–Cohn test for the *open* disc reads

$$|c| < 1 \quad \text{and} \quad |b - \bar{b}\,c| < 1 - |c|^2.$$

So, up to boundary cases, $\mathcal{P}_3$ is a semialgebraic set in $(a_2, a_3) \in \mathbb{C}^2$ described by these inequalities for all $x \in (0, 1)$. (A numerical cross-check of this description against the Dieudonné radius is reported in the appendix.)

Pitfalls

The closed-disc version is *not* just "replace $<$ by $\leq$": when $|c| = 1$ both roots must lie on the circle, which forces $b = \bar{b}c$ *and* $|b| \leq 2$. This degenerate case occurs exactly at the candidate: as $x \to 1$, $|c| \to 3|a_3|$, and S_3 has $a_3 = \frac{1}{3}$. Handle it by hand.

Two independent routes; do both.

1. *Exact (preferred).* Eliminate x (the condition must hold for all $x \in (0, 1)$, i.e. a quantifier) with a computer-algebra system: cylindrical algebraic decomposition or resultants in SageMath/Mathematica on the MacBook. Then maximize $|a_2|^2 + |a_3|^2$ on the resulting region by Lagrange multipliers. Rotation lets you take $a_2 \geq 0$, leaving 3 real variables.

2. *Certified branch-and-bound.* A priori box: $0 \leq a_2 \leq 2$ (Bieberbach) and $|a_3| \leq \frac{1}{3}$ (Lemma 1.4). For each box, either certify it lies outside $\mathcal{P}_3$ (some x violates the inequalities for every point of the box, via Arb intervals), or certify $1 + |a_2|^2 + |a_3|^2 \leq 2 - \delta$ on it. Boxes touching the candidate cannot be closed numerically; use a local second-order argument along $\partial\mathcal{P}_3$ there. Run on the MacBook first; move to a RunPod CPU pod if the box count explodes.

8. Step 6: $n = 3$ for all q, and certifying counterexamples

- *All q via majorization:* try to prove that $\log|S_3(e^{it})|$ majorizes $\log|p(e^{it})|$ for all $p \in \mathcal{P}_3$. Numerically test first (Step 3); a proof would give all q simultaneously.
- *Direct, q in compact ranges:* branch-and-bound in $(a_2, \operatorname{Re} a_3, \operatorname{Im} a_3, q)$ with rigorous enclosures of J_q (Arb; trapezoid rule with a proven error bound using analyticity of $|p|^q$ in a strip, Lemma 1.3). Treat $q \to 0$ and $q \to \infty$ separately.
- *Certifying a counterexample* (if Step 3 finds one): give exact (rational/algebraic) coefficients; prove univalence via a certified Dieudonné check (interval roots bounded away from $\mathbb{D}$, or explicit Schur–Cohn inequalities); compute J_q for both polynomials with rigorous error bounds.

9. Step 7: general n (research-level)

- $q = 2$: for real coefficients, univalence relates to nonnegativity of cosine polynomials (Fejér kernels; see Dimitrov et al.). Try to formulate $\max \sum a_k^2$ as an optimization over nonnegative trigonometric polynomials (a semidefinite program) and compare with S_n.
- All q: a polynomial analogue of the star-function/majorization argument.

10. Step 8: verification and write-up

1. Re-run all certificates from a clean checkout on a different machine (RunPod vs MacBook) and archive outputs with hashes.
2. Formalize the small lemmas (Lemmas 1.1–1.4) in Lean if Mathlib supports them; mainly an error check.
3. Have a faculty member (e.g. your complex analysis instructor) check the argument before posting.
4. Write the paper yourself; state precisely what is proved, what is numerical, and how AI tools were used.

Timeline at a glance

Step	Duration	Hardware
0 Literature	2–4 days	MacBook
1 Environment	1 day	MacBook, RunPod setup
2 Toolkit	2–3 days	MacBook
3 Exploration	1–2 weeks	MacBook ($n \le 4$), RunPod CPU ($n \ge 5$)
4 Structure	ongoing	MacBook + LLM API
5 $n = 3$, $q = 2$ proof	2–4 weeks	MacBook (CAS, Arb), RunPod if needed
6 $n = 3$, all q	open-ended	MacBook + RunPod CPU
7 General n	research	—

A. Numerical cross-check of the $\mathcal{P}_3$ description

Script: `crosscheck_p3.py` (delivered). For 3,000 random pairs (a_2, a_3) with $|a_2| \le 1.2$, $|a_3| \le 0.4$, we compared (A) the Schur–Cohn conditions of Step 5 on a grid of 3,000 values of $x \in (0, 1)$ with (B) the Dieudonné radius test $R(p) > 1$ on a grid of 3,000 values of θ. Points with $|R - 1| < 10^{-3}$ (4 of them) were skipped, since there grid error and the open/closed distinction matter.

Result (numerical): 2,996 agreements, 0 disagreements; 192 of the compared points were univalent and 2,804 were not. Also $R(z + \frac{1}{2}z^2) = 1$ and $R(S_3) = 1$ to 10^{-12}.

This supports the explicit description of $\mathcal{P}_3$ away from its boundary. It does *not* check the boundary itself, which is exactly where the extremal lives (see the pitfall box in Step 5).